Prototype

Clone weekly-meeting doc templates per team
Season 1 · GoF · Beginner

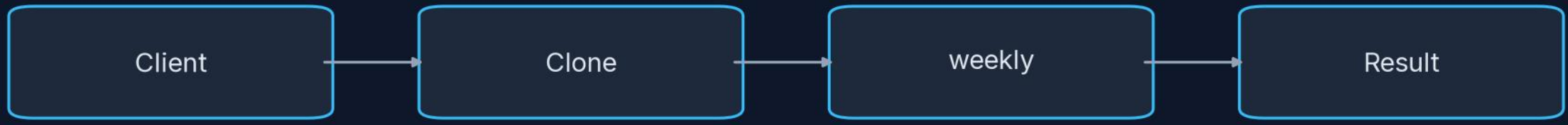
Today's Hook

- Problem: Clone weekly-meeting doc templates per team
- Naive pain cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- We will refactor to Prototype with a runnable Demo.
- Failure path is locked by ProblemTest.

The Problem

- Scenario: Clone weekly-meeting doc templates per team
- Constraints: solve «Clone weekly-meeting doc templates per team» with Prototype — keep the demo runnable and the failure path observable.
- Naive pain: Naive version fails to cleanly solve: Clone weekly-meeting doc templates per team
- Success looks like: Weekly (copy)

Structure



Prototype: Clone weekly-meeting doc templates per team

Real-World Scenario

- Running example: Clone weekly-meeting doc templates per team
- Key types: MeetingTemplate
- Happy-path stdout cue: Weekly (copy)
- Failure-path stdout cue: Failure path is asserted in ProblemTest (injected fault / invalid input)

Naive Code

```java
package com.buildwithwaqar.patterns.gof.creational.prototype.problem;

import java.util.*;

public final class NaiveTemplate {
  public static String clone(String title){ return title + " (copy)"; } // shallow, loses attendees
}
```

Why Naive Hurts

- Naive version fails to cleanly solve: Clone weekly-meeting doc templates per team
- Without Prototype, call sites absorb the wrong details.
- Failure mode to show on camera: Failure path is asserted in ProblemTest (injected fault / invalid input)
- New requirements force shotgun edits across the naive code.

Solution

```
package com.buildwithwaqar.patterns.gof.creational.prototype.problem;

import java.util.*;

import java.util.*;
final class MeetingTemplate implements Cloneable {
  private String title; private List<String> attendees;
  public MeetingTemplate(String title, List<String> attendees){
    this.title=title; this.attendees=new ArrayList<>(attendees);
  }
  public MeetingTemplate copy(){
    return new MeetingTemplate(title + " (copy)", new ArrayList<>(attendees));
  }
  public MeetingTemplate withTitle(String t){ this.title=t; return this; }
  public List<String> attendees(){ return List.copyOf(attendees); }
  public String title(){ return title; }
  public String describe(){ return title+":"+attendees; }
}
```

Runnable main

```
package com.buildwithwaqar.patterns.gof.creational.prototype.problem;

import java.util.*;

public final class ProblemDemo {
  public static void main(String[] args) {
    System.out.println("=== Prototype: clone meeting templates ===");
    System.out.println("--- naive ---");
System.out.println(NaiveTemplate.clone("Weekly"));
System.out.println("--- prototype ---");
var proto = new MeetingTemplate("Weekly Sync", List.of("Ada","Bob"));
var teamA = proto.copy().withTitle("Team A Sync");
System.out.println(proto.describe());
System.out.println(teamA.describe());
System.out.println("independent=" + (proto.attendees() != teamA.attendees()));
  }
}
```

Walkthrough Output

```
=== Prototype: clone meeting templates ===
--- naive ---
Weekly (copy)
--- prototype ---
Weekly Sync:[Ada, Bob]
Team A Sync:[Ada, Bob]
independent=true
```

Step-by-Step Trace

- Match stdout to stages — look for: Weekly (copy)
- Then the injected failure: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Re-run: java -cp … 07-prototype ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

Modern Java 25 / Zazr

```
// Java 25 direction for Prototype
// Prefer records + sealed where they shrink the demo.
// Problem: Clone weekly-meeting doc templates per team
```

Trade-offs

- Use Prototype when «Clone weekly-meeting doc templates per team» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind Prototype in product language.

Related Patterns

- Name the pattern usually confused with Prototype and separate the forces.
- Sketch the roles for «Clone weekly-meeting doc templates per team» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

Recap

- Problem: Clone weekly-meeting doc templates per team
- Naive pain → Prototype boundary → Demo stdout.
- Remember the failure cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Break a ProblemTest assertion, watch it fail, restore it.